

Mohammad Mavia

Backend Engineer · Web3 · AI · M.Sc. Cybersecurity Student

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Backend engineer with **five years building Web3 products**: wallets, a token launchpad with **0.5M+ registered investors**, DeFi lending, staking and bridges, and now perpetuals trading data for dYdX and HyperLiquid at Drox. I also own a production **multi-provider LLM pipeline**, and I wrote the plugin that lets an **LLM agent execute DeFi transactions** from a plain-language request. I care about the parts that have to be correct, not just fast. I am currently an **M.Sc. Cybersecurity student at BTU Cottbus–Senftenberg**, so much of what I build now sits where backend engineering meets security.

THINGS I HAVE BUILT

splitroute: Split-Route DEX Aggregation Engine (own project, Aug 2026)

Go · zero third-party dependencies · Prometheus metrics · Docker · GitHub Actions

- A swap router's real question is not which route is best but **whether to split the trade across several at once** once gas is priced in. I built the engine to answer it: constant-product and StableSwap pool maths in exact integers, bounded route search over a token graph, and three split allocators (greedy, dynamic programming, exhaustive) behind an HTTP API.
- Found and fixed the trap the obvious algorithm falls into. Greedy splitting is provably optimal without gas and I test it against exhaustive search over 200 random markets, but with gas it silently declines routes that would pay for themselves. It returned a worse trade in **218 of 400** sampled markets. On a 100k USDC swap the DP allocator opens a third route and delivers **+10.1 bps over the best single route**, against +5.1 bps for greedy.
- Chain state stays correct through reorgs. Each quote reads one immutable snapshot published through an atomic pointer, a reorg rewinds to a retained ancestor, and a reorg deeper than the retained window is **refused rather than guessed**. Profiling cut a quote from **2.88 ms to 0.87 ms**. About 2,900 lines of engine and 1,900 lines of tests (61 tests), run under the race detector in CI, with a decision document recording every design trade-off.

AI Agent for Automated DeFi Yield: On-chain Action Layer (Auto Compound, contract)

TypeScript · elizaOS-based agent · QuickSwap · The Graph · Polygon, BNB Chain

- The client's AI yield agent runs on a fork of elizaOS. I **authored the investment plugin** that lets it act on-chain: zapIn and zapOut of liquidity positions plus pool discovery, each with an LLM template that turns what the user typed into typed arguments for a signed on-chain transaction.
- Added QuickSwap and subgraph plugins so the agent reads real positions and history from The Graph instead of guessing, and the principal and APR maths across Polygon and BNB Chain pools, so the number the agent reports matches what the chain says. Delivered to a fixed deadline (92 of the repository's commits are mine).

Perpetuals Trading Data for dYdX v4 and HyperLiquid (at Drox)

TypeScript · NestJS · ClickHouse · WebSockets · HyperEVM

- The product profiles perpetuals traders from their on-chain activity, and I work on the data layer behind it: ingestion of trades, fills, funding and candles from dYdX v4 and HyperEVM/HyperLiquid, position reconstruction from raw fills, and per-user trade history with realised PnL aggregated from each fill.
- I wrote most of the user-facing service (a ~5,300-line trade, position and PnL service) and, alone, the WebSocket gateway that streams a long-running AI trader-profile job to the client as accepted, progress and result events instead of a request that hangs. **333 commits**, one of the repository's two main contributors, Oct 2025 to today.

ledger-core: Double-Entry Ledger Engine (own project, Sep 2026)

TypeScript on Node 22 · PostgreSQL 16 · one runtime dependency

- Anything that moves user money needs a record that cannot drift, so I built one: idempotent multi-leg transfers, two-phase holds, reversals as compensating entries, and integer amounts with no floating point anywhere. The balance rule is a deferred constraint in the database itself, so even a raw SQL write that bypasses the app cannot commit an unbalanced transfer. **22 tests**, including 400 overlapping transfers that conserve money exactly; ~1.25k transfers per second.

Production LLM Analysis Pipeline (at Drox)

Python · FastAPI · Anthropic · OpenAI · XGBoost

- Own the LLM layer of a crypto market-intelligence product: Claude as primary provider with GPT-4o as fallback, a separate circuit breaker per provider, and system prompts versioned v1 to v4 like code. The model explains numbers a deterministic precompute step produced, not numbers it made up. Part of a team codebase of ~124,000 lines of Python, where I am joint-largest contributor.

PROFESSIONAL EXPERIENCE

Drox

Jan 2024 – Present

Backend & AI Engineer

Remote

- Core engineer in a small, fully remote team with short decision paths, owning features end to end: API design, build, testing, rollout and on-call. Built the perpetuals data and LLM work above, and **multi-chain account abstraction** in the wallet server (transaction relaying, gas abstraction, signature validation) so users can transact without holding native gas on every chain.

Auto Compound

Jun 2025 – Aug 2025

Web3 & AI Engineer (Contract)

USA, Remote

- Delivered the AI agent’s on-chain action layer described above, from requirements to a running service.

BlockApex

Jun 2023 – Dec 2023

Backend Engineer (Part-time, alongside a full-time role)

Remote

- Built backend services and internal tooling for automated smart-contract analysis at a blockchain security firm. Proof that I can carry reduced hours next to other commitments.

Artfi

May 2023 – Nov 2023

Backend Engineer (Client-site)

Dubai, UAE

- A tokenised fractional-asset marketplace had no backend. Working **on-site with the client**, I designed and built it from zero, including the account onboarding flows (paper wallets and social login) that decide whether a non-crypto user ever gets in.

Xord

Oct 2020 – Jun 2023

Junior Fullstack Engineer → Software Engineer → Team Lead

Remote

- Promoted twice in under three years to **lead five engineers**, turning vague client ideas into technical designs and shipped products: wallets, staking, bridges, flash loans and lending on EVM chains, Solana, Near and Concordium.
- Architected the backend for a token launchpad with **0.5M+ registered investors** (cointerminal.com), using Merkle-root allowlists and Redis caching so the traffic spike at every sale opening never reached the database. Also built an Ethereum L2 mobile wallet on zkSync.

SKILLS

Languages	TypeScript/JavaScript, Python, Go, Solidity, SQL; Java; Rust in personal projects
Web3	EVM, Solana, dYdX v4, HyperEVM/HyperLiquid, zkSync and other L2s; ethers.js, Hardhat, ERC-20/721/1155, ERC-4337 account abstraction, The Graph subgraphs; DEX routing, liquidity-pool zaps, staking, lending, launchpads, perpetuals
AI	LLM agents with on-chain tools, Anthropic and OpenAI SDKs, versioned prompts, schema-constrained output, provider fallback; AI coding agents for daily prototyping
Backend & Data	NestJS, FastAPI, REST, WebSockets, gRPC, GraphQL; PostgreSQL, ClickHouse, MongoDB, Redis; RabbitMQ; Docker, Kubernetes, GitHub Actions, AWS, GCP

EDUCATION

M.Sc. Cybersecurity: currently enrolled

Nov 2025 – 2027 (expected)

Brandenburg University of Technology (BTU) Cottbus–Senftenberg, Germany

Certificate of enrolment available

B.Sc. Computer Science & IT

Sep 2017 – Jun 2021

NED University of Engineering & Technology, Pakistan

Also

Encode Solidity Bootcamp · Articles on Go and Rust concurrency and Go memory management (medium.com/@mauwia.atif) · English C1, Urdu native, German A1 and learning